



INVESTIGATIONS ON THE EFFECTS OF DC VOLTAGE CONTROL ON INERTIA PROVISION IN HVDC GRID-FORMING CONVERTER STATIONS

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AGENDA

- Background
- The improved GFM + DC voltage droop (DVD) control
- Small-signal tests of inertia in GFM Master stations and DVD stations
 - └ Constant RoCoF
 - └ Small-signal Frequency Oscillation (SFO)
- Performance comparisons in load jump transients
- Conclusions

BACKGROUND – THE LOW INERTIA CHALLENGES

- The challenges of frequency stability in future power systems
 - └ Synchronous machines replaced by power electronics converters – reduced rotating mass inertia.
 - └ Intermittency of wind / PV generations.
 - └ Susceptibility to high-RoCoF-induced SM disconnections, cascaded events, system split – reduced reliability and resiliency.

BACKGROUND – VIRTUAL INERTIA BY GFM

FNN Guideline: Grid forming behaviour of HVDC systems and DC-connected PPMs

Supplement to VDE-AR-N 4131 for dynamic frequency/active power behaviour and dynamic voltage control without reactive current specification



4-TSO Paper on Requirements for Grid-Forming Converters

6.2 HVDC systems

Active power exchange between the DUT and the network is generally required for complying to the requirement of dynamic frequency/active power behaviour and dynamic voltage control. HVDC systems practically have no inherent energy reserves (or are not commonly designed as such). Consequently, the associated energy balance requires sufficiently rapid compensation to ensure the stable operation of the HVDC system. Ideally, the source used for this energy compensation is independent of the AC system for which the functions of dynamic frequency/active power behaviour and dynamic voltage control are to be provided. Such an independent source may be, for example, another synchronous area connected via the DC circuit of the HVDC system.

Where energy compensation via an independent source is not feasible, the HVDC system can only provide its inherently available dynamic energy reserves. In this case, the energy balance has to be compensated within a few milliseconds by the affected AC system.

- GFM control could provide inertia response, BUT:
 - └ No inherent energy reserves in HVDC
 - └ Energy from other AC areas
 - └ Or limited energy supply during phase jump
 - └ How? Impact on DC voltage?

STATE OF THE ART

- How much inertia can a U-station (Master) provide while fulfilling the DC voltage requirements?

Grid Forming Control for HVDC Systems: Opportunities and Challenges

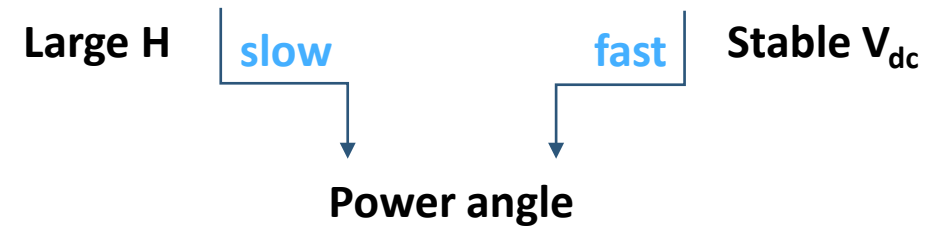
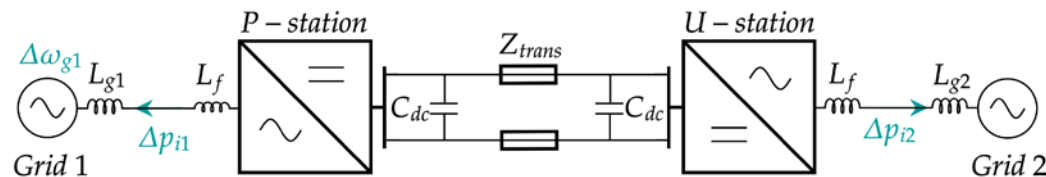
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Keywords

«Grid-forming control», «HVDC», «TSO»

Abstract

The aim of this paper is to highlight the opportunities and challenges of Grid Forming Control (GFC) for High Voltage Direct Current (HVDC) applications. A special focus will be on contradictory GFC performance requirements from European Transmission System Operators (TSOs). The impacts of these requirements on AC grid, HVDC performance and hardware are discussed.

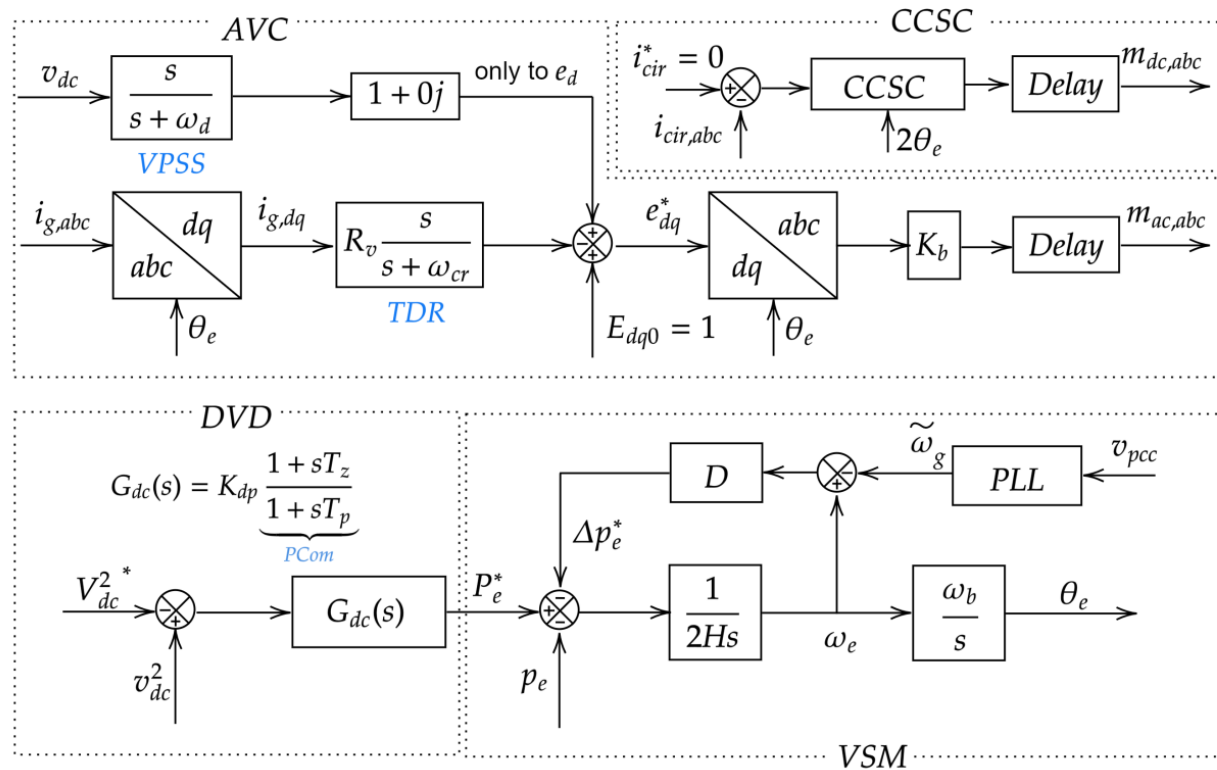


- There exists a maximum H for the U-station and that is the optimal design by trading off the frequency and DC voltage stability.
- Can the U-station truly provide inertia with virtual inertia?
- What if GFM is combined with DC voltage droop instead of single U-station?

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AN IMPROVED GFM + DVD FOR MTDC



Block diagram of the GFM control under study. [1]

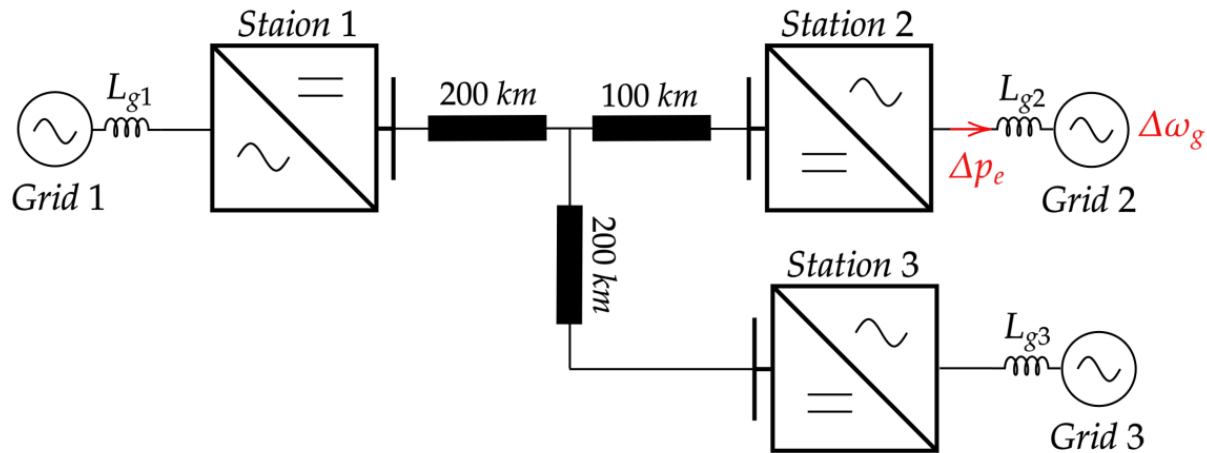
- The phase compensation with the DVD and the VPSS in AVC is added to enhance small-signal stability when combined with steep DVD gains needed for DC voltage security during N-1 contingency.
- G_{dc} is replaced by a PI control in the case of a master station.
- The same control structure is applied in all the study cases in this presentation.

[1] Y. Pang, A. Egea-Alvarez, J. C. Gonzalez-Torres, K. Shinoda, F. Perez and A. Benchaib, "DC Voltage Stability Analysis and Enhancement for Grid-Forming-Based MTDC Systems," in *IEEE Transactions on Power Electronics*

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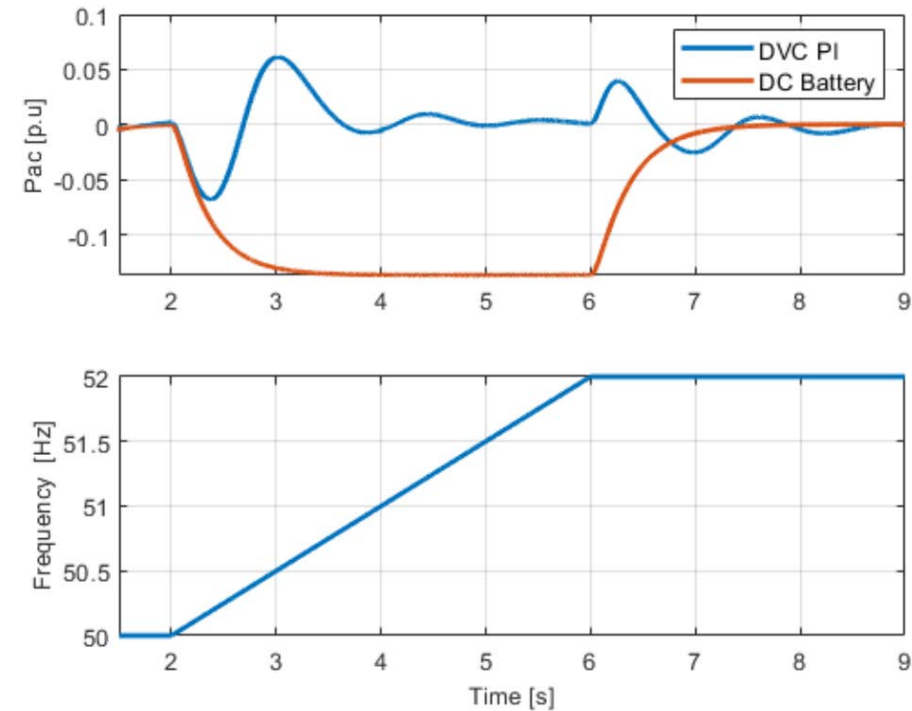
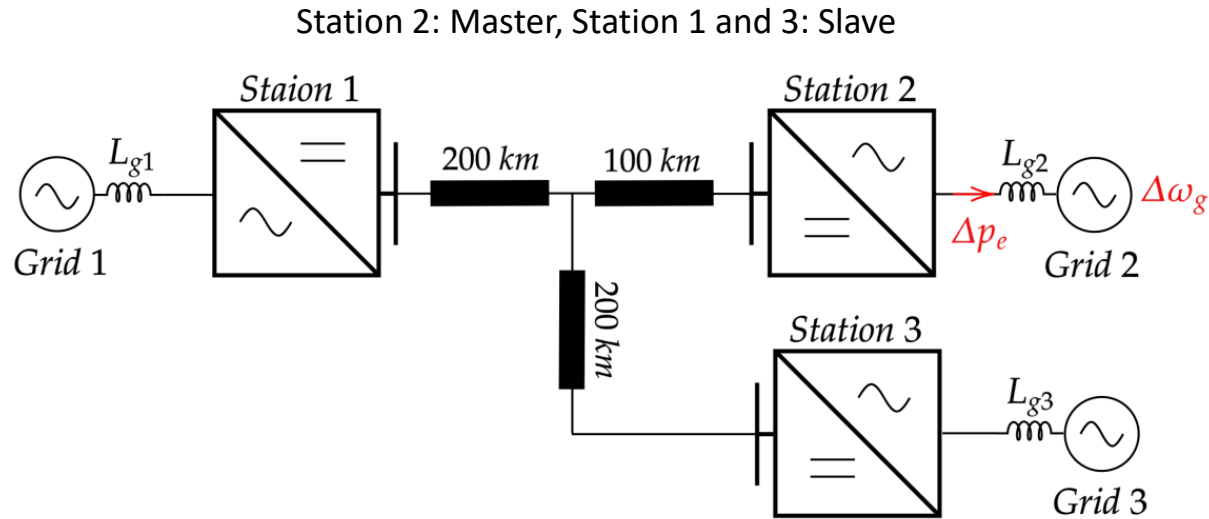
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A STUDY CASE OF A 3-TERMINAL HVDC GRID



- MMC model with non-energy-based CCSC + GFM in each station.
- DC cables models + DCRs are implemented on the DC side.
- AC side are represented by frequency-controllable ideal AC sources with grid impedances.
- ***Inject frequency disturbances and measure its active power response.***

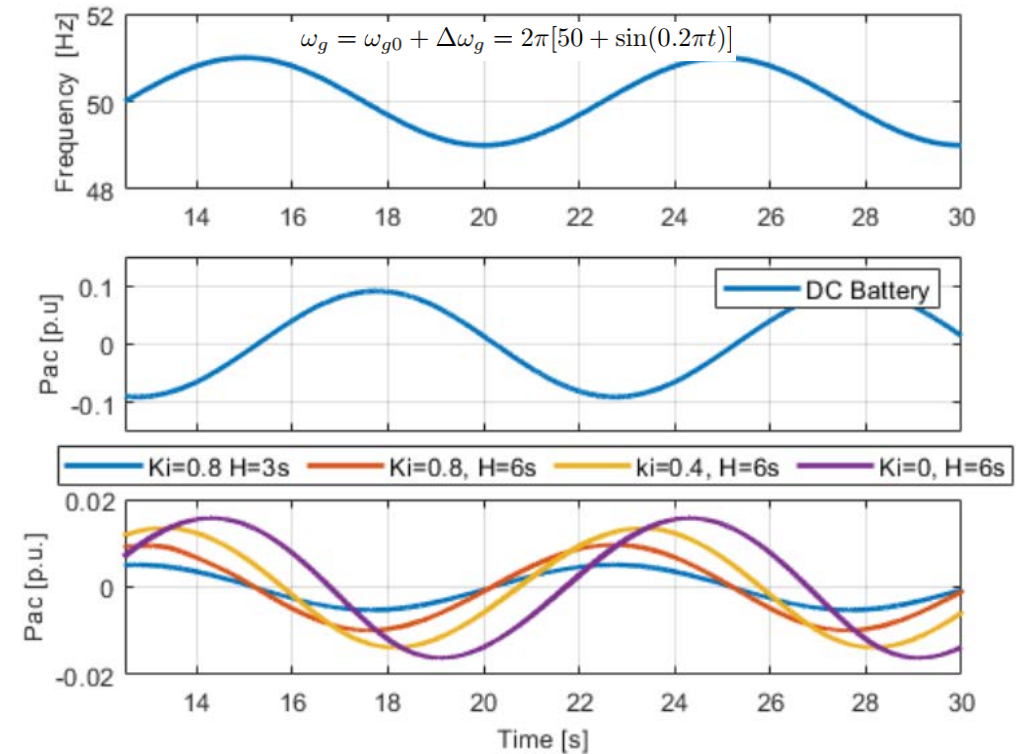
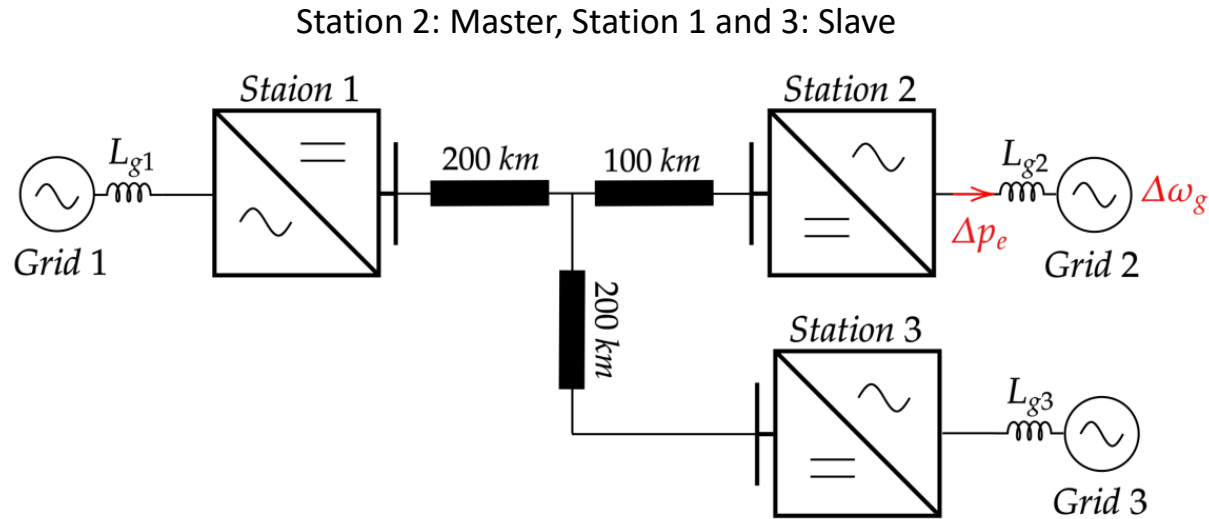
C-ROCOF TESTS AT GFM MASTER STATION



- Converter power response to constant RoCoF tests in EMT simulations. With the DC voltage control (DVC) based on PI controller added in the Master station, real inertia power response is completely inhibited.

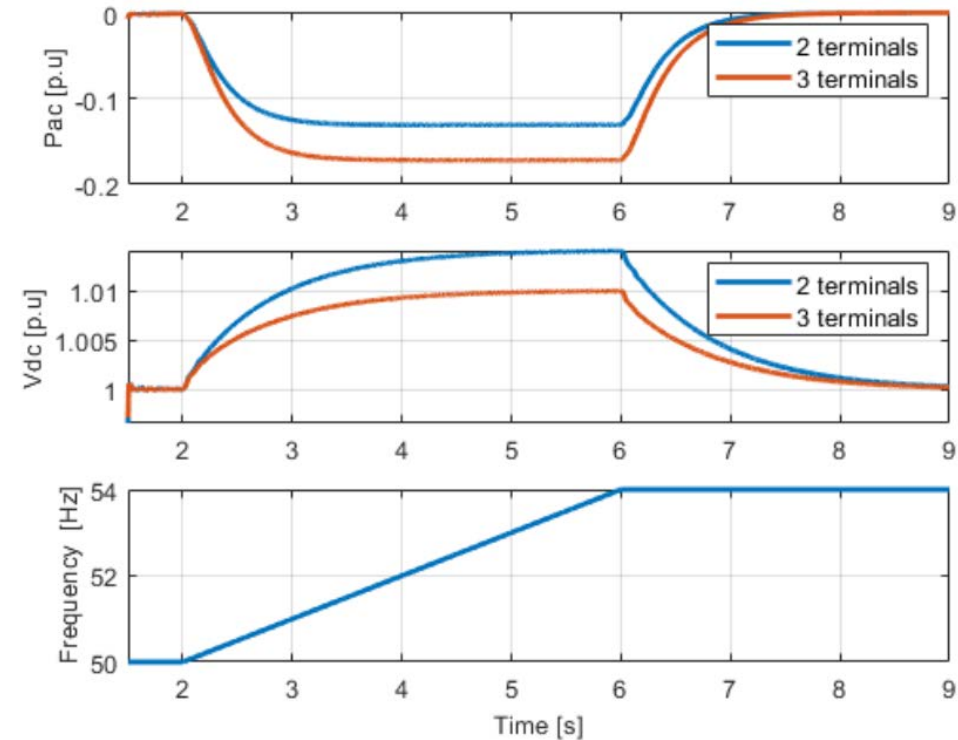
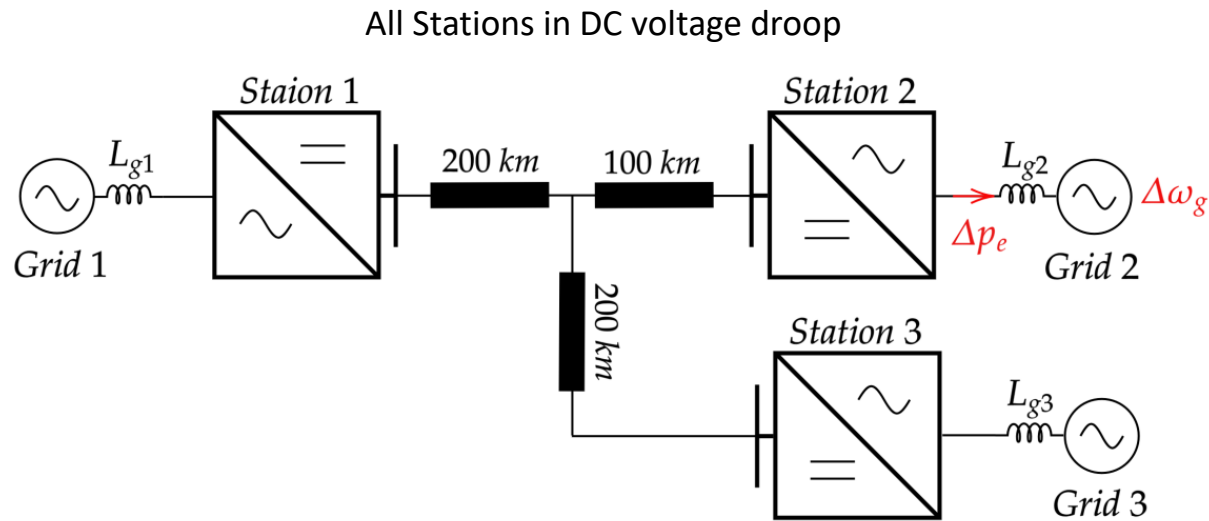
GC0137: Minimum Specification Required for Provision of GB Grid Forming (GBGF)

SFO TESTS AT GFM MASTER STATION



- With low-frequency SFO, the phase info of the power response is also revealed.
- Based on the phase, it can be observed that the Master GFM station is responding to the frequency oscillations with either **negative inertia or negative damping**, or a mixture of both, depending on the DVC PI parameters.
- **Increasing inertia constant in Master GFM station is detrimental to both AC and DC stabilities.**

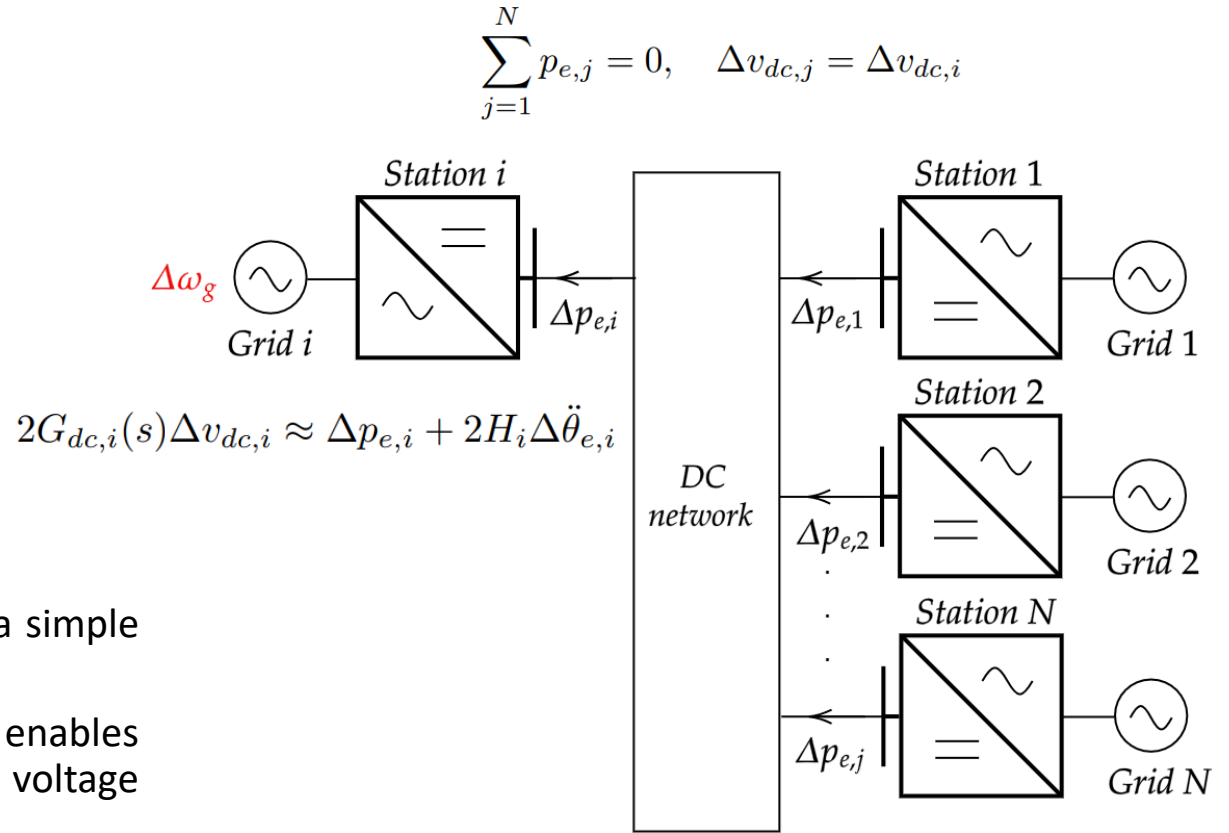
C-ROCOF TESTS AT GFM DVD STATION



- With more than one GFM stations in DC voltage droop, inertia response is enabled by dynamically dispatching power from other AC areas.
- The equivalent inertia scales up with the number of terminals in DVD and the DC voltage disturbance scales down.

HOW IS INERTIA FORMED BY DVD?

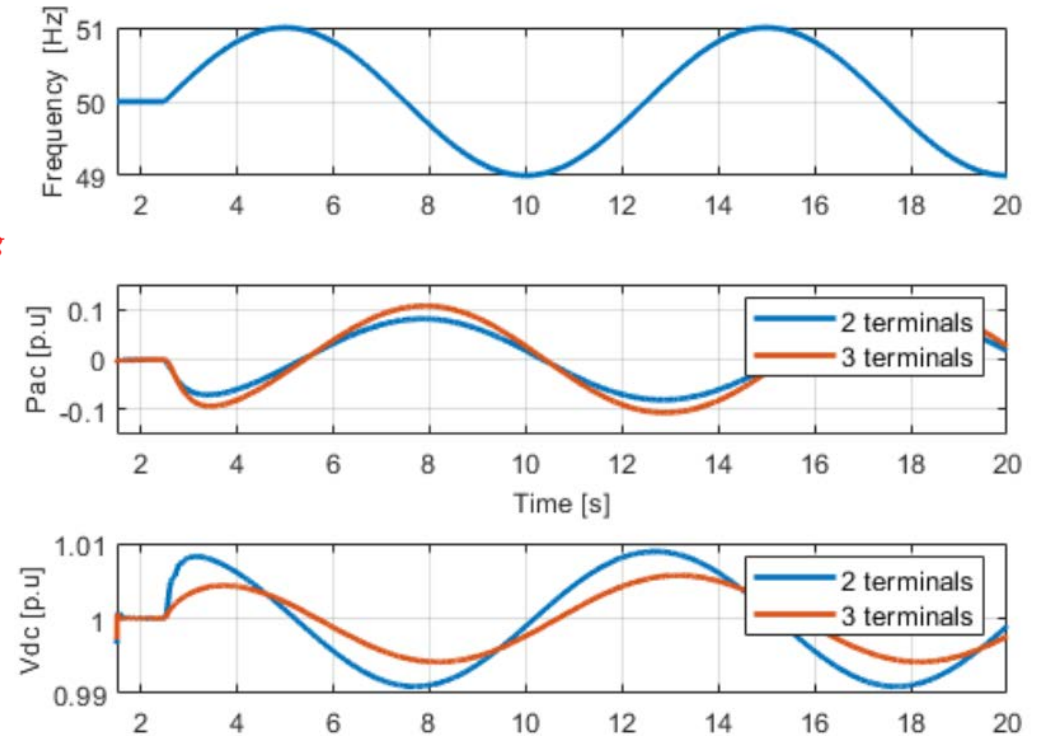
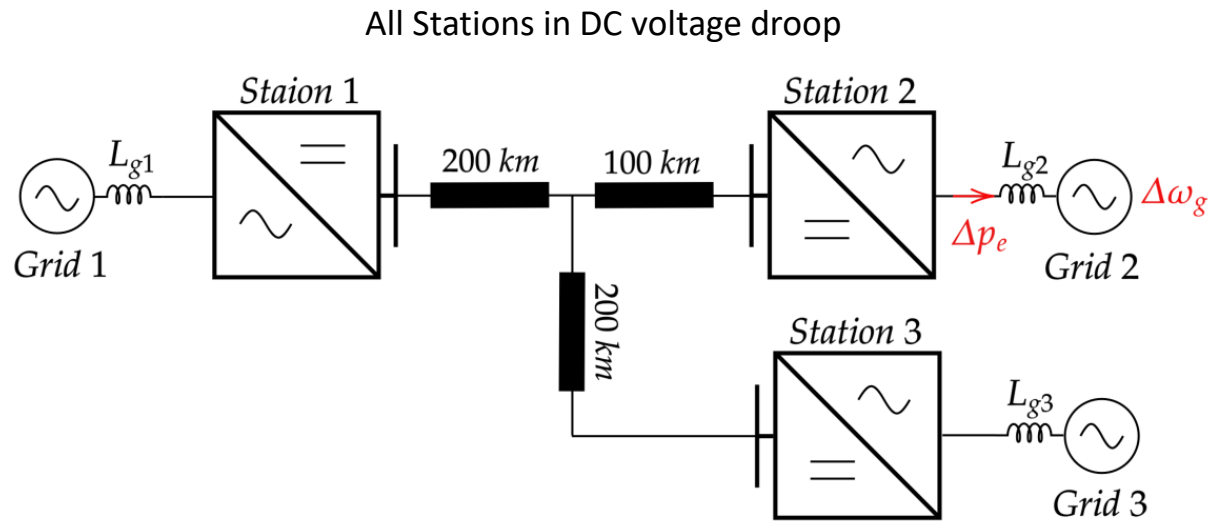
$$\begin{aligned}\Delta p_{e,i} &= -\frac{\sum_{j=1}^{j=N} G_{dc,j}(s) - G_{dc,i}(s)}{\sum_{j=1}^{j=N} G_{dc,j}(s)} 2H_i \Delta \ddot{\theta}_{e,i} \\ &= -\frac{\sum_{j=1}^{j=N} K_{dp,j} - K_{dp,i}}{\sum_{j=1}^{j=N} K_{dp,j}} 2H_i \Delta \ddot{\theta}_{e,i} \\ \Delta v_{dc,i} &= \frac{2H_i \Delta \ddot{\theta}_{e,i}}{\sum_{j=1}^{j=N} G_{dc,j}(s)} = \frac{2H_i \Delta \ddot{\theta}_{e,i}}{\sum_{j=1}^{j=N} K_{dp,j}(1 + sT_z)}\end{aligned}$$



- The effective inertia at one station is determined by a simple relations with the droop gains.
- Increasing the number of DVD stations in general enables more effective inertia power response with less DC voltage disturbances.
- The phase compensation in the improved DVD serves as a low-pass filter that reduces the DC voltage disturbances.

$$2G_{dc,j}(s)\Delta v_{dc,j} = \Delta p_{e,j}, j \in [1, N], j \neq i$$

SFO TESTS AT GFM DVD STATION

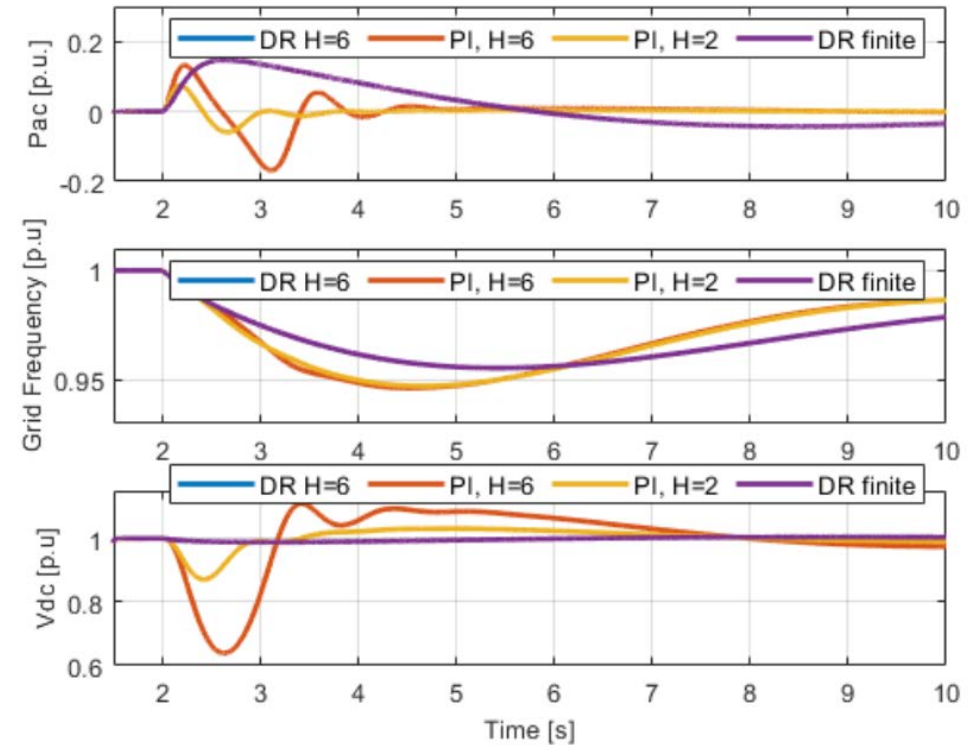
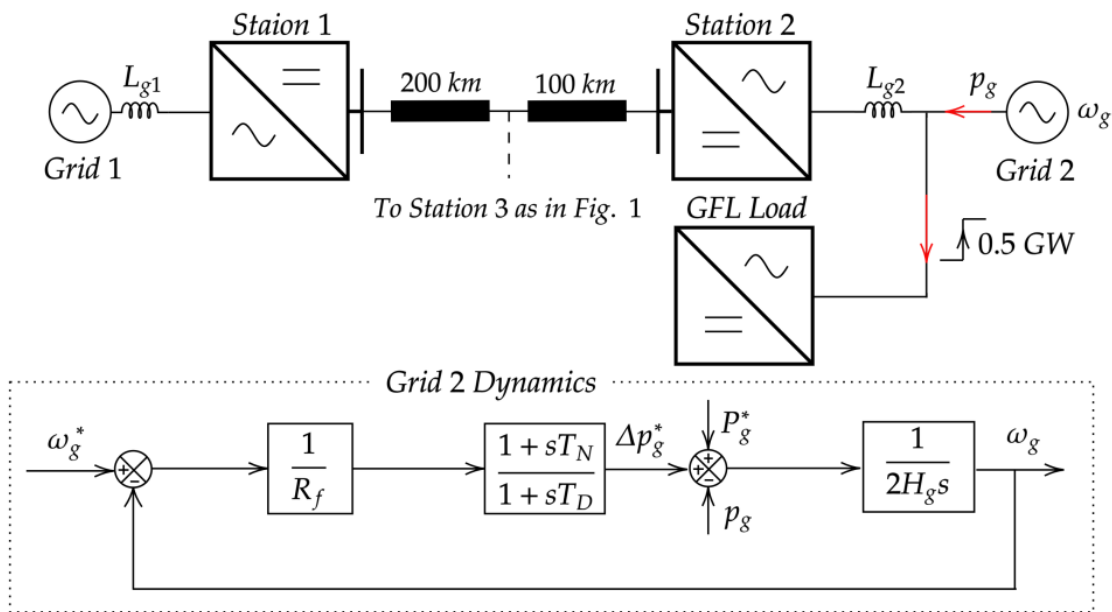


- Real inertia power response is also observed in SFO with GFM + DVD.
- The effective inertia, DC voltage disturbance follows the same relationship with the number of terminals and droop gains.

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PERFORMANCE COMPARISONS IN LOAD JUMP



- With Station 2 being the Master and Station 1 and 3 being the Slaves, increasing the inertia constant in Station 2 leads to slightly worst frequency nadir (negative inertia/damping effects) while strongly punishes the DC voltage stability.
- With all three stations in DVD, frequency nadir is improved due to actual inertia effects, with the DC voltage disturbance being negligible.
- Whether Grid 1 and 3 are infinite bus or not has little effects on the transient behaviors.

CONCLUSIONS

- ‡ No inertia can be provided by **GFM Master Station** but only **negative inertia / negative damping**. There exist **no trade-off** between frequency stability and DC voltage stability.
- ‡ **GFM Master Station** should only be expected to provide **voltage stiffness**.
- ‡ **GFM DVD Station** can provide **real inertia** power by dynamically dispatching power from other AC areas.
- ‡ Increasing the number of stations in DVD increases the effective inertia and reduces the DC voltage disturbance during AC transients.
- ‡ It should be noted that the improved GFM + DVD control is needed to secure the DC voltage stability.

THANK YOU!

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